

EMC WEEK- 12 PROJECT 2 BY VIGNESH S

1. **Create Operation:**

- Create a new database named "library" and switch to it.
- Create a collection named "books" with fields: title, author, and published_year.
- Insert a document into the "books" collection with the details of your favorite book.

```
> db.books.insertOne({
  title: "Your Favorite Book Title",
  author: "Author Name",
  published_year: 2020
})
< {
  acknowledged: true,
  insertedId: ObjectId('69f78edfa7252d9e1b1db48c')
}
```

2. **Read Operation:**

- Retrieve all documents from the "books" collection.

```
> db.books.find()
< {
  _id: ObjectId('69f78edfa7252d9e1b1db48c'),
  title: 'Your Favorite Book Title',
  author: 'Author Name',
  published_year: 2020
}
```

```
> _MONGOSH
> db.books.insertOne({
  title: "Your Favorite Book Title",
  author: "Author Name",
  published_year: 2020
})
< {
  acknowledged: true,
  insertedId: ObjectId('69f78edfa7252d9e1b1db48c')
}
> db.books.find()
< {
  _id: ObjectId('69f78edfa7252d9e1b1db48c'),
  title: 'Your Favorite Book Title',
  author: 'Author Name',
  published_year: 2020
}
> db.books.find({ author: "J.K. Rowling" })
<
> db.books.insertOne({
  title: "Harry Potter",
  author: "J.K. Rowling",
  published_year: 1997
})
< {
  acknowledged: true,
  insertedId: ObjectId('69f78fc7a7252d9e1b1db48d')
}
> db.books.find({ author: "J.K. Rowling" })
```

```
_MONGOSH
author: 'Author Name',
published_year: 2020
}
> db.books.find({ author: "J.K. Rowling" })
<
> db.books.insertOne({
  title: "Harry Potter",
  author: "J.K. Rowling",
  published_year: 1997
})
< {
  acknowledged: true,
  insertedId: ObjectId('69f78fc7a7252d9e1b1db48d')
}
> db.books.find({ author: "J.K. Rowling" })
< {
  _id: ObjectId('69f78fc7a7252d9e1b1db48d'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  published_year: 1997
}
> db.books.find().sort({ published_year: 1 }).limit(1)
< {
  _id: ObjectId('69f78fc7a7252d9e1b1db48d'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  published_year: 1997
}
library>
```

Exercise 3 – Update Operation:

1. Update the year for “The Catcher in the Rye” (if it exists, otherwise insert it first):

```
> db.books.updateOne(
  { title: "The Catcher in the Rye" },
  { $set: { published_year: 2026 } }
)
db.books.find({ title: "The Catcher in the Rye" })
<
> db.books.updateMany(
  {},
  { $set: { genre: "Mystery" } }
)
db.books.find()
< {
  _id: ObjectId('69f78edfa7252d9e1b1db48c'),
  title: 'Your Favorite Book Title',
  author: 'Author Name',
  published_year: 2020,
  genre: 'Mystery'
}
{
  _id: ObjectId('69f78fc7a7252d9e1b1db48d'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  published_year: 1997,
  genre: 'Mystery'
}
library>
```

Exercise 4 – Delete Operation.

4(a) Remove document with title "1984"

```
db.books.insertOne({
  title: "1984",
  author: "George Orwell",
  published_year: 1949,
  genre: "Mystery"
})
{
  acknowledged: true,
  insertedId: ObjectId('69f790c0a7252d9e1b1db48e')
}
library>|
```

```
> db.books.deleteOne({ title: "1984" })
db.books.find()
< {
  _id: ObjectId('69f78edfa7252d9e1b1db48c'),
  title: 'Your Favorite Book Title',
  author: 'Author Name',
  published_year: 2020,
  genre: 'Mystery'
}
{
  _id: ObjectId('69f78fc7a7252d9e1b1db48d'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  published_year: 1997,
  genre: 'Mystery'
}
library>|
```

4(b) Delete all books where published_year < 2000

```
> db.books.deleteMany({ published_year: { $lt: 2000 } })
db.books.find()
< {
  _id: ObjectId('69f78edfa7252d9e1b1db48c'),
  title: 'Your Favorite Book Title',
  author: 'Author Name',
  published_year: 2020,
  genre: 'Mystery'
}
library>|
```

5. **Advanced Query:**

```
> use library
db.books.find().sort({ published_year: -1 }).limit(3)
< already on db library
> db.books.insertMany([
  { title: "MongoDB Basics", author: "John Doe", published_year: 2023, genre: "Tech" },
  { title: "Learning NoSQL", author: "Jane Smith", published_year: 2022, genre: "Tech" }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f791b7a7252d9e1b1db48f'),
    '1': ObjectId('69f791b7a7252d9e1b1db490')
  }
}
library>|
```

```
> db.books.find({
  $or: [
    { title: /MongoDB/i },
    { title: /NoSQL/i }
  ]
})
< {
  _id: ObjectId('69f791b7a7252d9e1b1db48f'),
  title: 'MongoDB Basics',
  author: 'John Doe',
  published_year: 2023,
  genre: 'Tech'
}
{
  _id: ObjectId('69f791b7a7252d9e1b1db490'),
  title: 'Learning NoSQL',
  author: 'Jane Smith',
  published_year: 2022,
  genre: 'Tech'
}
library>
```

```
,
> // Type 2: only MongoDB
db.books.find({ title: /MongoDB/i })

// Type 3: only NoSQL
db.books.find({ title: /NoSQL/i })
< {
  _id: ObjectId('69f791b7a7252d9e1b1db490'),
  title: 'Learning NoSQL',
  author: 'Jane Smith',
  published_year: 2022,
  genre: 'Tech'
}
library>
```